Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Nanoparticle-enabled Radical Scavenging for Chemically Durable Proton Exchange Membrane Fuel Cells			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	Fuel Cells	Fine Chemicals	Nano-Application Technologies
Concept of Proposed Technology	The proposed technology is a nanoparticle-enabled membrane modification platform designed to enhance the chemical durability of proton exchange membranes (PEMs) for hydrogen fuel cell systems, targeting radical-induced membrane degradation as a primary failure mechanism during long-term operation[1,2]. Conventional approaches based on metal-ion infusion or filler incorporation suffer from scavenger mobility, aggregation, and limited positional control, leading to insufficient durability improvement[3,4]. To address these limitations, the proposed approach employs radical scavenging nanoparticles with precisely engineered surface properties, which are stably immobilized at interfacial regions within the membrane architecture. The nanoscale dimensions of the scavengers provide high effective surface area and fine spatial control, enabling localized radical suppression while limiting unintended perturbation of the membrane's intrinsic proton transport characteristics[5,6]. Integrated with scalable membrane fabrication strategies, the platform establishes a practical foundation for long-lifetime, high-durability PEMs in next-generation hydrogen energy systems.			
Rationale for Joint R&D Project	Advancing nanoparticle-enabled durability enhancement strategies for proton exchange membranes (PEMs) from materials concepts to TRL 4 and beyond requires coordinated consideration of nanoscale interfacial design, polymer electrolyte membrane chemistry, and implementation strategies aligned with fuel cell operating conditions[7,8]. In particular, sustained radical mitigation in PEMs under chemically and mechanically demanding environments depends critically on the stability and spatial control of nanoparticle–polymer interfaces, necessitating an integrated academic–industrial approach[9].			

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	The academic team (RPI) combines expertise in nanoparticle design, self-assembled surface modification, and nanoscale interface engineering (Ji-Young Kim, Ph.D.) with complementary strengths in PEM-specific polymer chemistry, membrane processing, and structure–property relationships for fuel cell applications (Chul Sung Bae, Ph.D.)[10-16]. This integrated academic capability enables the rational design of interfacially immobilized radical scavenging nanostructures compatible with advanced PEM materials and long-term chemical durability requirements[17]. In parallel, the Korean SME leads on membrane-scale implementation, process-compatible integration of nanoparticle-based modifications, and performance evaluation under fuel cell–relevant operating conditions. By addressing practical considerations such as manufacturability, operational robustness, and reproducibility, this academic–industrial collaboration establishes a clear pathway for translating nanoscale interfacial design principles into industrially viable, high-durability PEM technologies for next-generation hydrogen energy systems[18].
Key Words	1. Fuel cell 2. Durability 3. Proton exchange membrane 4. Nanoparticle–polymer interface engineering

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level (2-3)	**End of the Project**	Level (4-5)
R&D Objectives	Proton exchange membrane fuel cells (PEMFCs) are increasingly deployed in transportation, stationary power, and energy conversion systems; however, their long-term reliability remains constrained by chemical degradation of proton exchange membranes (PEMs) under extended operating conditions. During fuel cell operation, reactive radical species continuously attack membrane polymers, leading to fluoride release, membrane thinning, increased gas crossover, and progressive performance decay, which collectively limit PEMFC operational lifetime[19]. Conventional approaches to enhance PEM chemical durability, including the use of ionic additives or inorganic fillers, have been investigated, but these strategies often suffer from additive mobility, non-uniform distribution within the membrane, long-term stability degradation, or disruption of proton transport pathways, restricting their effectiveness under practical fuel cell conditions[20,21]. The objective of this research is to develop a composite PEM platform based on surface-functionalized radical scavenging nanoparticles that suppresses radical-induced chemical degradation while maintaining membrane performance required for fuel cell operation. This will be achieved by establishing nanoparticle–PEM interfacial design and immobilization strategies that suppress nanoparticle aggregation and migration during membrane fabrication and fuel cell operation, thereby enabling sustained radical scavenging functionality within the membrane architecture[22,23]. In parallel, the effects of nanoparticle incorporation on membrane structure and proton transport properties will be evaluated to define the performance window associated with durability enhancement. This research further aims to verify that chemical durability improvements achieved at the membrane level are retained at the membrane electrode assembly (MEA) and fuel cell device levels. By correlating membrane-level durability metrics with MEA-level performance and degradation behavior, the applicability and scalability of the proposed composite PEM platform for practical fuel cell systems will be assessed [24-28].			

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R&D Target Specifications	The final target is to realize a composite PEM with well-dispersed radical scavengers capable of stable operation, low background interference, and data-driven signal interpretation. More detailed specifications are followings: 1. Chemical Durability Enhancement of Composite PEMs [29-32] This target focuses on achieving substantial improvement in PEM chemical durability through stabilized radical scavenging functionality. By immobilizing surface-functionalized radical scavenging nanoparticles within the membrane architecture, localized suppression of radical-induced degradation will be achieved without inducing nanoparticle migration or aggregation during fabrication and operation. Chemical durability will be quantitatively evaluated using standardized metrics relevant to PEM fuel cell operation, including open-circuit voltage (OCV) decay rate and fluoride emission rate (FER). Successful achievement of this target will demonstrate that nanoparticle-enabled radical mitigation can effectively suppress membrane degradation mechanisms responsible for long-term performance decay. *Target metrics:* • Open-circuit voltage(OCV) decay rate ≤ 0.4 mV h^{-1} • Fluoride emission rate (FER) ≤ 0.5 μg F cm^{-2} h^{-1} 2. Preservation of Proton Transport Performance[33] This target aims to ensure that chemical durability enhancement does not compromise proton transport, addressing a key limitation of conventional durability strategies. Through rational nanoparticle–polymer interfacial design, the proposed approach seeks to preserve interconnected proton conduction pathways and hydrated ionic domain structures within the membrane. Proton conductivity will be evaluated under fuel cell–relevant conditions to verify that stabilized nanoparticle incorporation mitigates the conventional trade-off between chemical durability and ion transport performance. *Target metrics:* • Proton conductivity ≥ 0.09 S cm^{-1} at 80 °C and 100% RH 3. Stability and Immobilization of Radical Scavenging Nanoparticles This target focuses on achieving long-term stability of radical scavenging functionality through controlled nanoparticle immobilization within PEM matrices. Surface functionalization strategies will be designed to enhance interfacial affinity between nanoparticles and PEM polymers, thereby suppressing aggregation during membrane fabrication and preventing nanoparticle migration under chemically and mechanically demanding fuel cell operating conditions. Stability will be evaluated at the formulation, membrane, and durability-testing levels to confirm that radical scavenging functionality is retained throughout processing and operation. *Target metrics:* • No detectable nanoparticle migration across membrane thickness after accelerated chemical durability testing (cross-sectional SEM/EDS or STEM-EDS mapping) • ≤ 20% change in nanoparticle dispersion characteristics following membrane fabrication and accelerated stress testing (TEM/STEM imaging) • ≥ 80% retention of radical scavenging activity after accelerated chemical durability testing (ex situ radical scavenging assay pre/post testing) 4. Membrane-to-Device Translation and MEA-Level Validation This target aims to validate the practical relevance of the developed composite PEMs through membrane electrode assembly (MEA) fabrication and fuel cell testing. By correlating membrane-level chemical durability metrics with MEA-level performance and degradation behavior, this target ensures that durability enhancements achieved at the materials and membrane levels are preserved at the

	device scale. Successful achievement of this target will demonstrate the readiness of the nanoparticle-enabled PEM durability strategy for translation toward industrial fuel cell systems. *Target metrics:* • ≥ 30–50% reduction in MEA-level OCV decay rate compared to reference PEM-based MEAs • ≥ 90% retention of initial electrochemical performance following durability testing
R&D Plan	**Aim 1. Initiation of RPI–Korean SME collaboration and definition of nanoparticle-enabled PEM durability strategies** (0–6 months) This aim focuses on initiating the collaboration between the PIs at RPI and a Korean SME and establishing a shared technical and implementation framework for nanoparticle-enabled chemical durability enhancement in proton exchange membranes (PEMs). We will define representative PEM material systems, durability metrics, and operating conditions relevant to fuel cell applications, and will establish initial design strategies for surface-functionalized radical scavenging nanoparticles and their immobilization within PEM architectures. In parallel, a concrete plan for intellectual property management and joint development pathways will be formulated. These efforts will culminate in a unified development roadmap guiding subsequent material, membrane, and device-level studies. **1.1. Selection of target PEM systems and durability evaluation metrics** (Kim–Bae–Korean SME) Representative PEM material systems will be selected to reflect state-of-the-art fuel cell membranes and emerging chemically durable PEM candidates. Target operating conditions, including temperature, humidity, and electrochemical stress environments, will be defined in alignment with practical PEMFC operation. Key chemical durability metrics, such as fluoride emission rate (FER) and open-circuit voltage (OCV) decay rate, will be identified and standardized to ensure consistent evaluation across material, membrane, and MEA levels. **1.2. Definition of nanoparticle-enabled radical scavenging and immobilization strategies** (Kim–Korean SME) Initial design strategies for radical scavenging nanoparticles will be established, with emphasis on surface functionalization approaches that promote strong interfacial affinity with PEM polymers while suppressing aggregation and migration. Candidate nanoparticle classes and interfacial chemistries will be defined based on their potential to enable localized radical suppression without perturbing proton transport pathways. These strategies will be evaluated in the context of compatibility with membrane fabrication and scalability considerations. **Aim 2. Engineering of surface-functionalized radical scavenging nanoparticles for PEM integration** (0–12 months) This aim focuses on the synthesis and optimization of radical scavenging nanoparticles with precisely engineered surface chemistries to enable stable dispersion and immobilization within PEM matrices. Surface functionalization parameters will be systematically varied to tune nanoparticle–polymer interactions and to define a controllable interfacial design space suitable for membrane fabrication. **2.1. Design and synthesis of radical scavenging nanoparticles with controlled surface chemistry** (Kim) Radical scavenging nanoparticles will be synthesized and surface-functionalized using controllable chemical modification routes. The degree and nature of surface functionalization will be systematically varied to modulate nanoparticle–polymer interfacial interactions, with the objectives of suppressing agglomeration and enhancing compatibility with PEM polymers. Intrinsic radical scavenging activity will be evaluated through ex situ chemical degradation assays. **2.2. Evaluation of nanoparticle–PEM compatibility and formulation-level stability** (Kim–Bae) Formulation-level dispersion stability and interfacial compatibility between surface-functionalized nanoparticles and PEM polymers will be evaluated using microscopic and spectroscopic characterization techniques. The influence of surface chemistry on dispersion stability, aggregation

tendencies, and potential impacts on ionomer morphology will be assessed to guide down-selection of optimized nanoparticle candidates.

2.3. Assessment of scalability and process compatibility (Korean SME)
Surface functionalization strategies and nanoparticle synthesis routes will be assessed from the perspective of scalability, reproducibility, and compatibility with membrane fabrication workflows. Feedback from this assessment will inform refinement of nanoparticle design toward industrially viable formulations.

Aim 3. Fabrication and membrane-level validation of chemically durable composite PEMs (7–24 months)

This aim integrates optimized surface-functionalized nanoparticles into PEMs and evaluates chemical durability and proton transport performance at the membrane level. Nanoparticle loading and degree of surface functionalization will be treated as key formulation variables to establish structure–property relationships governing durability and conductivity.

3.1. Fabrication of composite PEMs incorporating immobilized radical scavenging nanoparticles (Kim)
Composite PEMs will be fabricated using processing routes compatible with conventional membrane manufacturing. Nanoparticle incorporation strategies will be optimized to ensure uniform distribution and stable immobilization within the membrane architecture. Processing parameters will be adjusted to maintain membrane integrity and reproducibility.
3.2. Chemical durability and proton transport performance evaluation of composite PEMs (Kim-Bae-Korean SME)
Chemical durability of the composite membranes will be quantitatively evaluated using standardized metrics, including fluoride emission rate (FER) and accelerated chemical degradation tests. The impact of nanoparticle loading and surface functionalization on suppression of radical-induced degradation will be systematically analyzed. Proton conductivity will be measured under fuel cell–relevant conditions (e.g., 80 °C, 100% RH). The role of nanoparticle immobilization and interfacial design in preserving interconnected proton conduction pathways will be examined, with performance benchmarked against reference PEMs.

Aim 4. Long-term stability assessment and MEA-level validation under fuel cell operating conditions (19–30 months)

This aim prepares the developed composite PEMs for practical fuel cell implementation by addressing long-term operational stability and device-level performance.

4.1. MEA fabrication using optimized composite PEMs (Kim-Bae–Korean SME)
Selected composite membranes will be integrated into membrane electrode assemblies (MEAs) using established fabrication protocols. MEA fabrication parameters will be optimized to ensure reliable device assembly and performance consistency.

4.2. MEA-level durability and performance evaluation (Kim-Bae–Korean SME)
MEA-level durability will be evaluated through open-circuit voltage (OCV) decay measurements and accelerated stress testing under PEMFC operating conditions. Correlations between membrane-level chemical durability and MEA-level performance will be established to validate translation across length scales.

Aim 5. Optimization and generalization of nanoparticle-enabled PEM durability platforms (25–36 months)

This aim focuses on refining nanoparticle surface chemistry, membrane formulation, and processing parameters to maximize robustness, reproducibility, and generality across PEM systems.

5.1. System-level optimization of nanoparticle–PEM architectures (Kim–Bae)

	Interfacial design parameters will be refined to improve reproducibility and performance consistency across multiple PEM formulations. General design rules linking nanoparticle surface chemistry to durability and transport performance will be established. **5.2. Assessment of industrial readiness and future scale-up potential** (Korean SME) The developed nanoparticle-enabled membrane modification strategies will be evaluated for manufacturability, scalability, and alignment with industrial PEM production workflows, positioning the technology for future TRL advancement beyond the current project.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute assumes responsibility for establishing the scientific foundation, materials design principles, and membrane-level validation framework underlying the proposed nanoparticle-enabled PEM durability platform. Ji-Young Kim, Ph.D., is responsible for the design, synthesis, and surface functionalization of radical scavenging nanoparticles, as well as the development of nanoparticle-based interfacial strategies for chemical durability enhancement in proton exchange membranes. The Kim group brings core expertise in nanoscale material synthesis, chemical modification, and controlled self-assembly of nanoparticles, enabling precise control over nanoparticle size, surface chemistry, dispersion behavior, and immobilization within polymer matrices. In addition, the group investigates nanoparticle–polymer interfacial interactions and their influence on local physicochemical environments relevant to radical suppression and membrane stability, supported by advanced structural characterization and nanoscale property analysis. Chulsung Bae, Ph.D., is responsible for providing expertise in polymer electrolyte membrane selection, modification, and processing to support the integration of surface-functionalized radical scavenging nanoparticles. The Bae group contributes knowledge in ionomer chemistry, membrane morphology, and polymer–additive compatibility, assisting in the identification of suitable PEM systems and in the modification of membrane formulations or processing conditions as needed to enable stable nanoparticle incorporation. This includes evaluating how nanoparticle incorporation influences membrane structure, ion-conducting domain organization, and chemical stability under fuel cell–relevant environments. Together, the Kim and Bae groups are responsible for defining materials design principles, membrane-level performance benchmarks, and durability validation protocols for the composite PEM platform. Through close collaboration, the two groups integrate nanoscale nanoparticle and interface design with polymer electrolyte selection and membrane processing expertise to establish structure–property relationships governing chemical durability and transport behavior.

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Expected R&R of Korean SME Partner	The Korean SME partner is responsible for translating the nanoparticle-enabled PEM durability concepts established by the research institute into practically implementable membrane fabrication and process-relevant strategies. The SME's primary responsibilities focus on process compatibility assessment, composite PEM manufacturing support, and evaluation under application-oriented requirements relevant to fuel cell deployment. First, the Korean SME will work in close collaboration with the research institute to support the process-level implementation of composite PEM architectures incorporating surface-functionalized radical scavenging nanoparticles. This includes evaluating nanoparticle incorporation routes, dispersion processes, and membrane formation conditions in the context of existing PEM manufacturing workflows, with attention to mechanical integrity, handling properties, and long-term stability under fuel cell operating environments. Second, the SME will take the lead in assessing process reproducibility, manufacturing robustness, and scalability considerations. This involves evaluating batch-to-batch consistency, sensitivity of composite PEM properties to processing parameters, and feasibility of scaling nanoparticle functionalization and membrane fabrication steps. Based on these assessments, the SME will provide feedback to refine materials design and processing strategies toward industrial relevance. In addition, the Korean SME will define membrane-level requirements and system-interface constraints relevant to fuel cell integration, ensuring that the developed composite PEMs are compatible with downstream membrane–electrode assembly (MEA) fabrication and fuel cell system requirements. This role focuses on identifying practical constraints and performance windows rather than conducting device-level electrochemical testing. Through these activities, the Korean SME partner will support the maturation of the nanoparticle-enabled PEM durability platform beyond laboratory-scale proof-of-concept toward a technology base with demonstrated process feasibility and industrial applicability, consistent with advancement toward TRL 4 and beyond.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This research is expected to generate meaningful academic and industrial impact by establishing nanoparticle-enabled composite proton exchange membrane (PEM) durability enhancement as a core materials design strategy for fuel cell technologies. By implementing radical scavenging functionality in a robust and broadly applicable manner within the membrane architecture, the proposed approach effectively mitigates chemical degradation mechanisms that have long limited the operational lifetime of PEM fuel cells, thereby providing a technical foundation for long-lifetime and high-reliability fuel cell systems. From an academic perspective, this project will systematically elucidate the relationships between interfacial stabilization and immobilization behavior of surface-functionalized nanoparticles and membrane-level chemical durability. By complementing and extending conventional durability enhancement approaches based on ionic additives or inorganic fillers, the study establishes a new materials design framework for functionalized polymer electrolyte membranes. In particular, the project quantitatively links nanoparticle surface chemistry, dispersion state, and interfacial interactions to radical-induced degradation suppression and membrane structural stability, thereby providing conceptual and methodological foundations for the design of polymer electrolyte membranes incorporating functional nanoparticles. These outcomes are expected to be broadly applicable not only to next-generation PEM architectures but also to a wide range of electrochemical energy systems where long-term stability of functional additives is critical. From an industrial perspective, the proposed research directly addresses one of the key bottlenecks limiting PEM fuel cell commercialization, insufficient chemical durability of the membrane under prolonged operating conditions. The nanoparticle-enabled durability enhancement strategy is

	designed to be highly compatible with existing PEM manufacturing processes, enabling implementation with minimal process modification. As a result, the proposed approach is expected to contribute to extended system lifetime, improved performance stability, and reduced maintenance costs, generating tangible technological and economic value across transportation, stationary power generation, and hydrogen energy infrastructure applications. Furthermore, the composite PEM modification strategy developed in this project, based on modular design principles and process compatibility, offers long-term flexibility for potential technology extension or collaboration with overseas companies and research institutions engaged in fuel cell development. In this way, the project primarily supports the strengthening of domestic fuel cell technology competitiveness, while also providing a foundation for future international dissemination and application.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization